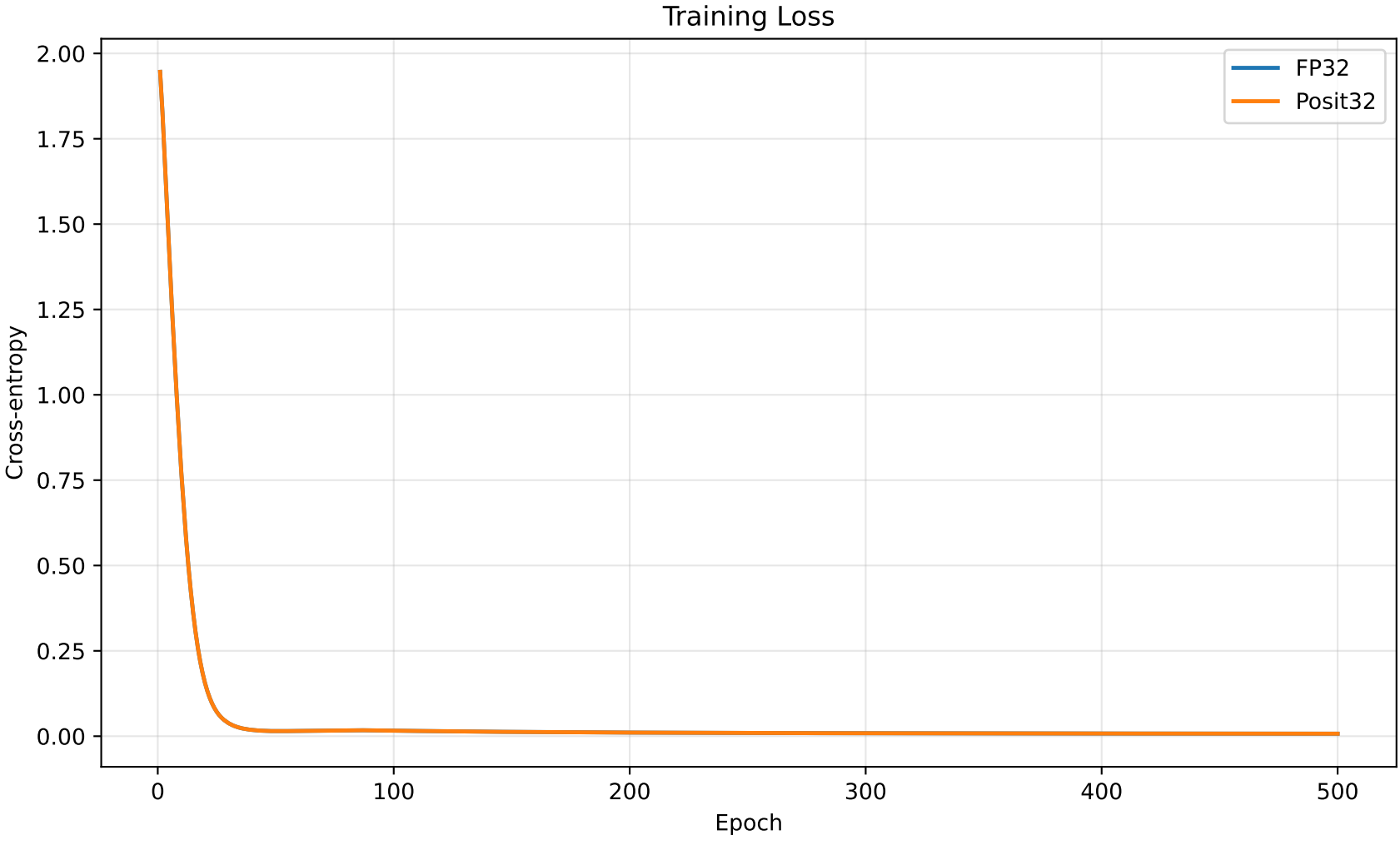
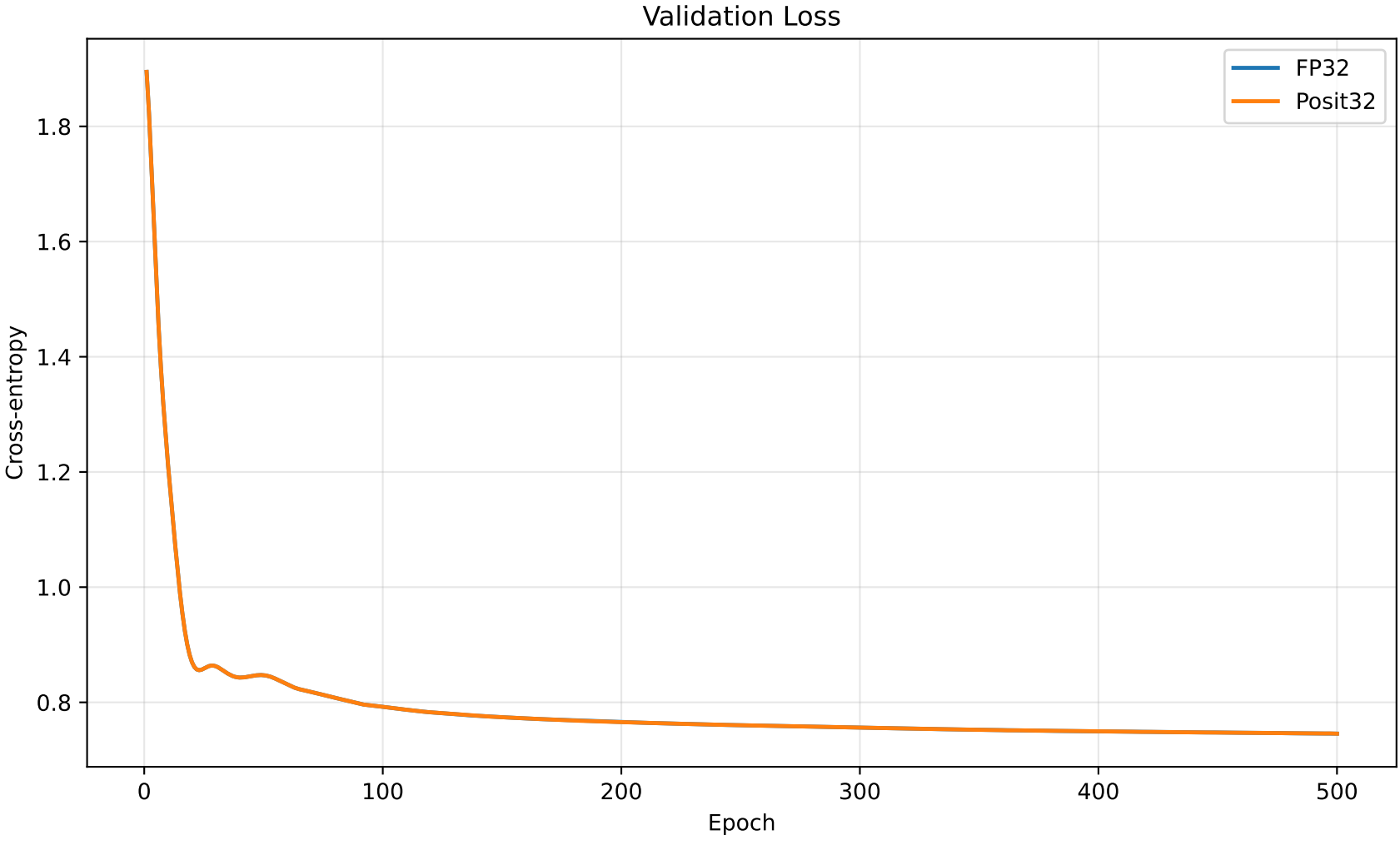
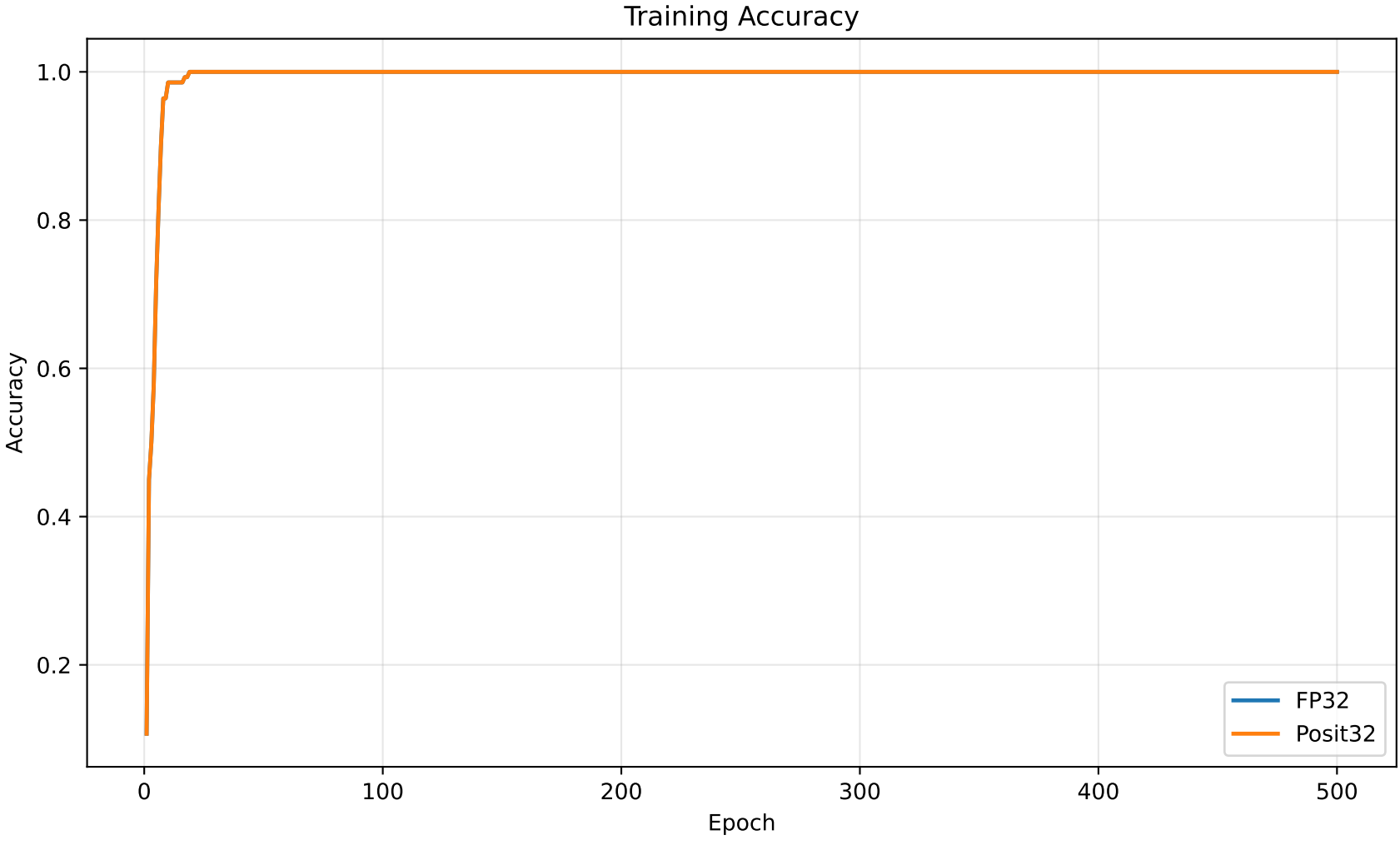
FP32 vs Posit32 GCN — 500-Epoch Adam

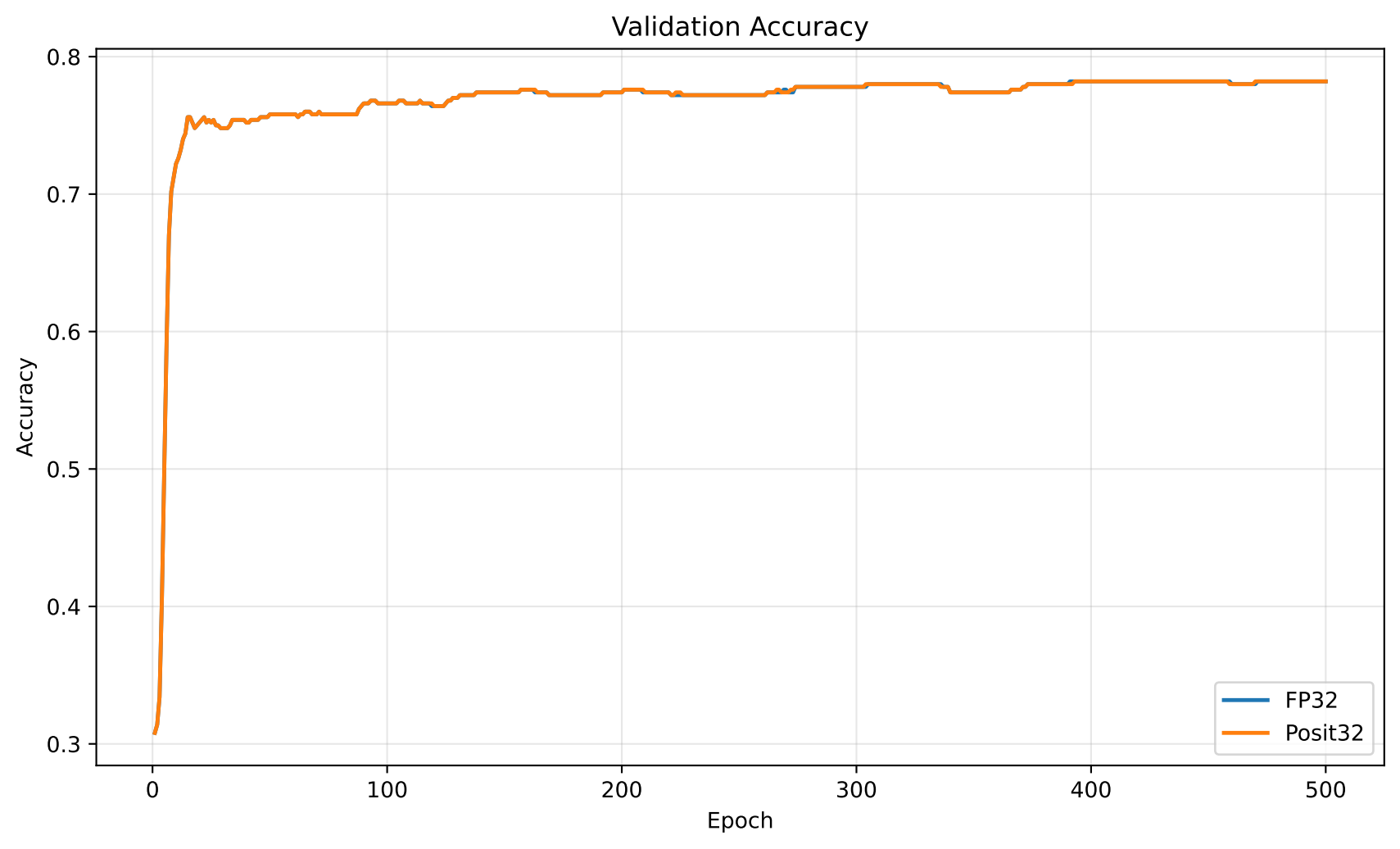
Metric	FP32	Posit32
Training accuracy	100.0%	100.0%
Validation accuracy	78.2%	78.2%
Test accuracy	80.4% (804/1000)	80.4% (804/1000)
Training cross-entropy	0.00730936	0.00730484
Validation cross-entropy	0.745627	0.745556
Test cross-entropy	0.627936	0.627886

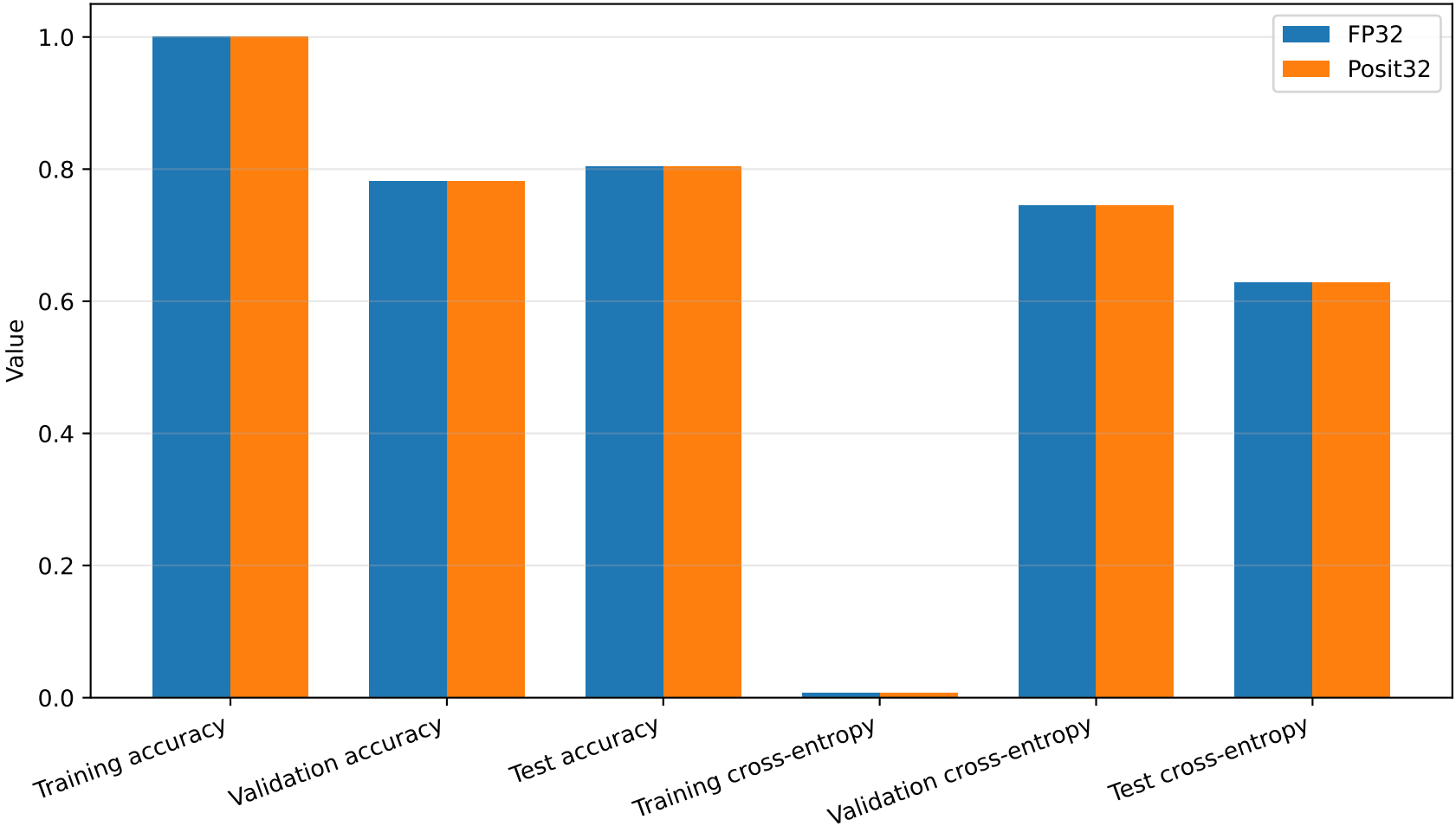
Same evaluation procedure; saved checkpoints; no training updates during evaluation.


Training Loss
FP32
Posit32
Cross-entropy
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
100
200
300
400
500
Epoch


Validation Loss
FP32
Posit32
Cross-entropy
1.8
1.6
1.4
1.2
1.0
0.8
0
100
200
300
400
500
Epoch


Training Accuracy
Accuracy
Epoch
1.0
0.8
0.6
0.4
0.2
0
100
200
300
400
500
FP32
Posit32


Validation Accuracy
Accuracy
Epoch
0.8
0.7
0.6
0.5
0.4
0.3
0
100
200
300
400
500
FP32
Posit32


Final Checkpoint Metrics: FP32 vs Posit32
FP32
Posit32
Value
1.0
0.8
0.6
0.4
0.2
0.0
Training accuracy
Validation accuracy
Test accuracy
Training cross-entropy
Validation cross-entropy
Test cross-entropy